

Uday Tyagi

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Education

Cornell University – Master's in Computer Science | GPA 3.8/4.0 Dec. 2025 – Present

University of Wisconsin–Madison – B.S. in Computer Science | GPA 3.8/4.0 May 2023 – Dec. 2025

Coursework: Advanced ML Systems, Neural Networks, Algorithms, Operating Systems, Deep Learning, Robot Learning, Computer Vision

Research

Researcher – LAISR (Long-Term AI Safety Research) · Cornell (Prof. Lionel Levine) Jan. 2026 – Present

- Constitutional Drift: measuring whether an LLM's stated values remain stable when the model repeatedly rewrites its own constitution, tracking semantic drift, value collapse, and self-reinforcing edits across successive rounds of self-revision.
- Built the experimental harness with two collaborators: multi-round self-editing pipelines across model families, embedding- and judge-based drift metrics, and behavioral evals testing whether edited constitutions actually change downstream model behavior.

AI Safety CAMBRIA Program – Cambridge Boston Alignment Initiative May – June 2026

- Worked through the ARENA mechanistic interpretability curriculum at CBAI's 3-week intensive: linear probes, steering vectors, activation patching, sparse autoencoders, OthelloGPT, emergent misalignment, alignment faking, and LLM evals.
- Original research using Natural Language Autoencoders (Fraser-Taliente et al., 2026): generated effective steering vectors for Qwen2.5-7B from hand-written text alone, with no labeled data, no activation extraction, and no fine-tuning; vectors successfully shift model behavior across concepts (sycophancy, misalignment) and are geometrically concept-specific. Writeup in progress for LessWrong.

Research Engineer – NASA USRC (Project ORION) · Cornell University Jan. 2026 – Present

- Trained a diffusion-based motion planner replacing ORCA for real-time multi-drone collision avoidance: DiT architecture generating velocity command sequences from ego and neighbor states, with DPM-Solver++ inference optimization and low-temperature reranking; reached ~80% mean scenario completion vs. ORCA's 59%.
- Built the surrounding deconfliction stack: ROS 2 planner plugin, scenario-based evaluation pipelines with behavioral analytics, and a ~200k-trajectory training dataset across six collision scenarios.

AI Safety Research Fellow – AI Safety Camp Dec. 2025 – April 2026

- Evaluated hierarchical and parallel AI control protocols on the ControlArena benchmark, measuring safety-usefulness tradeoffs as the capability gap between trusted and untrusted models grows.
- Extended ControlArena with red-team/blue-team protocols and Elo-based scoring to measure failure thresholds of AI oversight mechanisms under adversarial prompting.

Researcher – SMARAG, Cornell (Prof. Oliver Gao Lab) Jan. 2026 – Present

- Built a multi-agent RAG system for automated ESG carbon reporting (LangChain orchestration, RL for factual grounding); designed an agent observability layer with audit-trail logging and behavioral guardrails to detect hallucinated citations in regulatory filings.

Experience

Machine Learning & Software Engineer – KLA · Remote May 2025 – Present

- Fine-tuned an LLM-powered AI agent for the DART web application, enabling app engineers to interact with defect inspection workflows in natural language, from image classification to event routing and triage.
- Built visual regression and UI-performance monitoring pipelines for HPC automation workflows using Grafana Faro and custom metrics tooling, reducing debugging time in large-scale wafer inspection systems.

Software Engineering Intern – Foresee Health · San Francisco, CA Sept. 2024 – Jan. 2025

- Engineered an authentication system for the invoice application using AWS Amplify and Lambda; designed PostgreSQL schemas and synchronization patterns for a microservices architecture handling large customer datasets.

Projects

NLA Steering: Text-Driven Behavioral Steering of LLMs

Python, PyTorch, HuggingFace, Qwen2.5-7B

- Used Natural Language Autoencoders (Fraser-Taliente et al., 2026) to generate steering vectors for Qwen2.5-7B from hand-written text descriptions alone — no training data, no activation extraction, no labeled examples; the resulting vectors successfully and reliably shift model behavior at inference time across multiple concepts (sycophancy, misalignment, persona).
- Running follow-on experiments to understand what makes this work: ablating which parts of the text format carry the steering signal, iterative refinement through the autoencoder loop, and benchmarking the vectors as zero-shot behavioral probes against ground-truth activation-derived vectors.

Emergent Misalignment: Reproduction, Probing & Activation Oracles

Python, PyTorch, HuggingFace, PEFT, LoRA

- Fine-tuned Qwen2.5 on a small set of harmful financial advice examples and reproduced emergent misalignment — broad misaligned behavior generalizing to unrelated prompts; implemented a custom LoRA layer (SimpleLoraLinear) from scratch, verified numerical equivalence with HuggingFace PEFT, and trained linear probes on intermediate LoRA activations to detect misalignment signals before generation.
- Ran activation oracles on the misaligned model: decoded residual-stream activations verbally through an Activation Verbalizer to detect and characterize misalignment in natural language directly from internal states — a training-free, interpretable complement to the linear probes.

Domain-Adaptive LLM Embedding Benchmark & Fine-Tuning

Python, PyTorch, HuggingFace, sentence-transformers

- Generated semantically adversarial “hard negative” passages via LLM prompting and used them as contrastive training data to fine-tune a retrieval model, improving retrieval accuracy across scientific and code domains; benchmarked 5+ embedding models and introduced a robustness metric quantifying how often each is fooled by near-miss passages.

LLM Jailbreak Defense Evaluation

Python, PyTorch, HuggingFace

- Engineered jailbreak experiments combining GCG and AutoDAN automated attacks with layered defenses (self-reminders, hierarchical prompting, perplexity filters); evaluated guard-model architectures via Attack Success Rate metrics.

Wisconsin School Analytics on GCP

BigQuery, Dataform, GCS, SQL, Python

- Built a cloud-native analytics pipeline integrating geospatial and public school datasets across 400+ schools; automated Parquet ingestion via PyArrow with parameterized SQL on remote VMs.

Leadership

Operations Lead – Cornell AI Alignment (CAIA)

Jan. 2026 – Present

- Lead operations and semester planning for CAIA: coordinating fellowships, running the weekly technical paper reading group, and connecting students with AI safety research opportunities.

Fundamentals Cohort Lead – Wisconsin AI Safety Initiative

Sept. 2024 – Dec. 2025

- Led a 10-student cohort through the AI Safety Fundamentals curriculum, facilitating weekly technical discussions on alignment, interpretability, and AI governance.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, Go, SQL, CUDA

AI/ML: PyTorch, HuggingFace, mechanistic interpretability (probes, steering vectors, SAEs, activation patching), fine-tuning (LoRA/PEFT), evals, RAG, diffusion models, LangChain, inference optimization

Tools & Infra: Docker, Kubernetes, AWS (S3, Lambda), GCP, ROS 2, PostgreSQL, Kafka, Spark, React, Node.js